

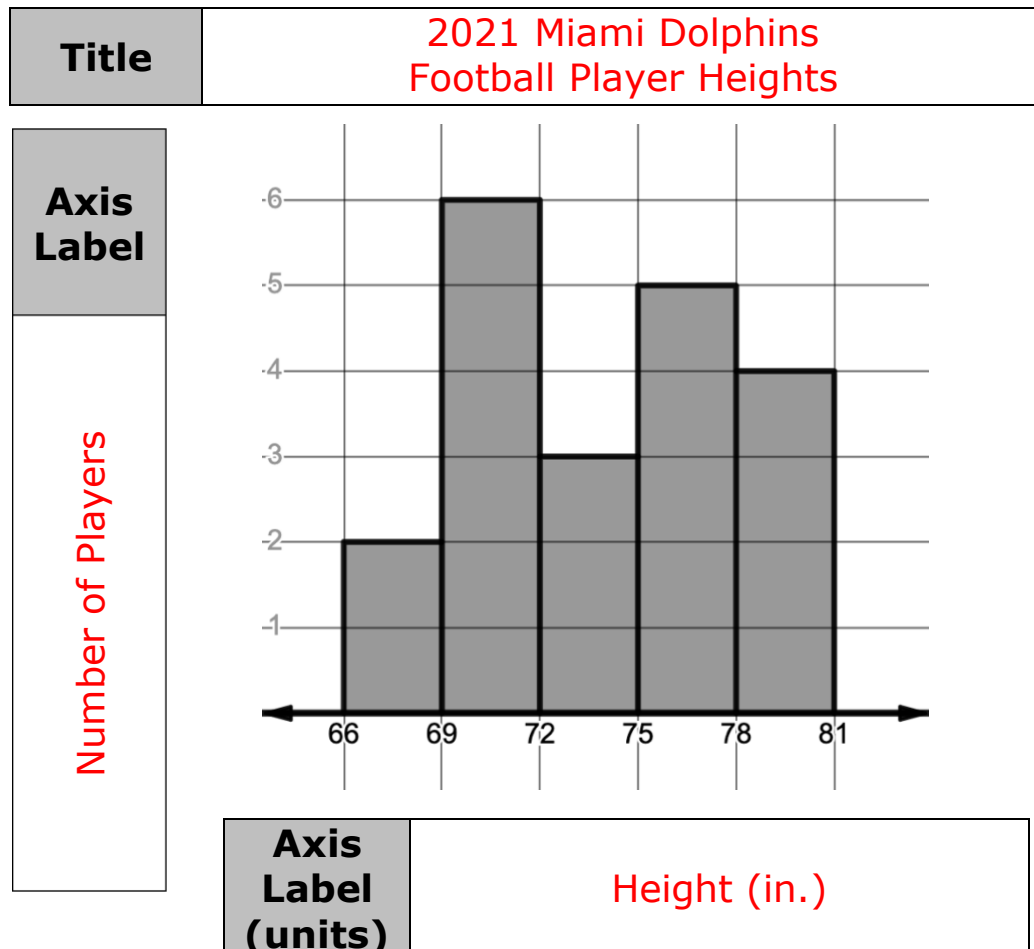
Showing Work with a Calculator: Because you are using a calculator, it is possible to accidentally punch the wrong button. To ensure you earn partial credit in each work box show the operation(s) that you entered into the calculator and the result.

1. The height, in inches (in.), of some of the Miami Dolphins football players during the 2021 season is shown.

71, 71, 69, 70, 70, 78, 77, 77, 77, 77, 75, 78, 68, 79, 78, 68, 73, 71, 73, 73

An incomplete histogram of the data is shown. Complete the histogram by providing:

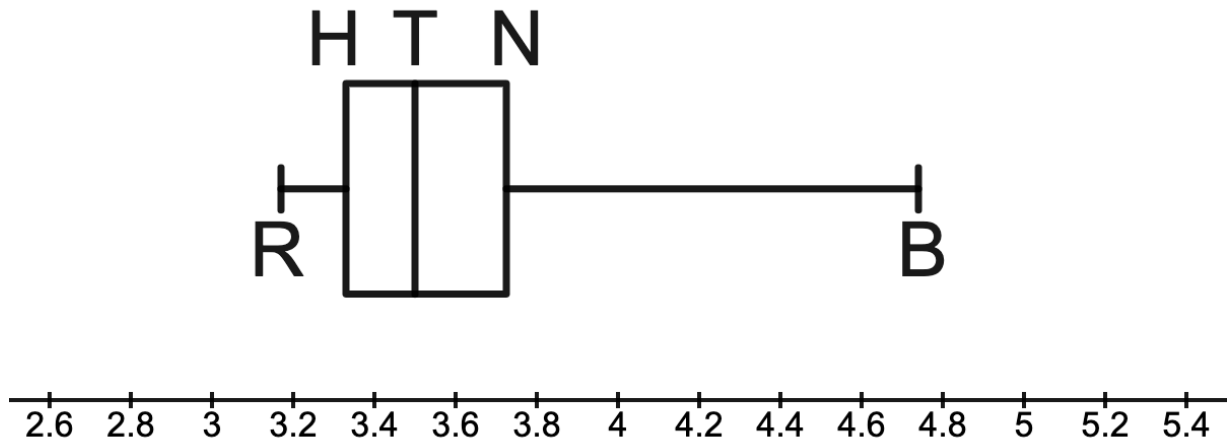
- a title
- a label for the vertical axis
- a label for the horizontal axis.



2. The US Bureau of Labor Statistics tracks the price of a pound of chocolate chip cookies. Some of the prices are shown.

$\{\$3.19, \$3.50, \$3.17, \$3.50, \$3.48, \$3.55, \$3.47, \$3.90, \$4.74\}$

The box plot of the data is shown.



- a. Complete the table by providing the value for each label.

Label	Value	Label	Value
B	4.74	T	3.5
H	3.33	R	3.17
N	3.725		

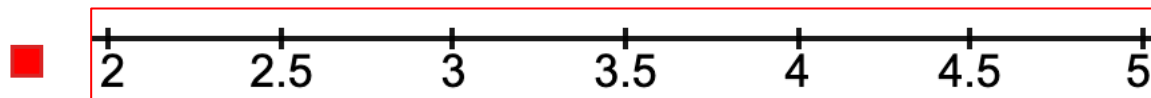
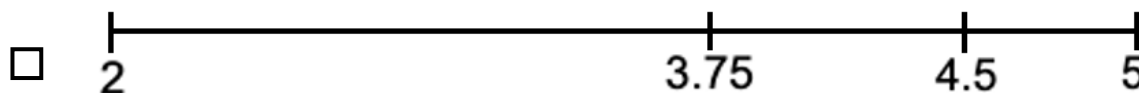
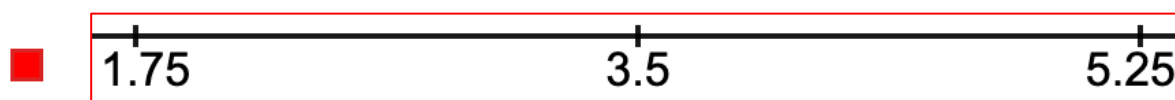
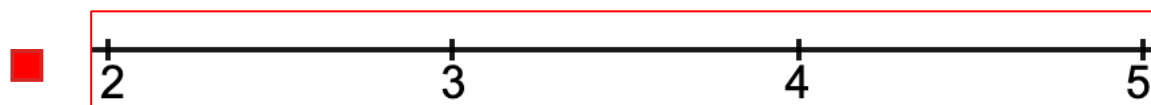
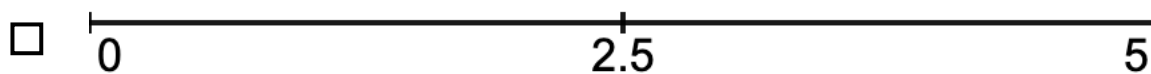
- b. Select all of the statements that are true.

- ☐ 10% of the data are between H and T .
- ☒ 25% of the data are between T and N .
- ☒ 25% of the data are between R and H .
- ☒ 50% of the data are between H and N .
- ☐ 50% of the data are between R and N .
- ☐ 75% of the data are between N and B .

3. Kadeesha designs clothes. The table shows the amount of fabric, in yards (yd), she needs to buy.

Design Name	Fabric (yd)			
	Fabric A	Fabric B	Fabric C	Fabric D
Caheerah	$3\frac{1}{4}$	$2\frac{3}{4}$	$3\frac{1}{2}$	$4\frac{1}{8}$
Hania	$4\frac{1}{8}$	$3\frac{1}{2}$	$2\frac{1}{4}$	$2\frac{3}{4}$
Pallaton	$3\frac{3}{4}$	$4\frac{1}{8}$	$4\frac{3}{4}$	$3\frac{5}{8}$
Reilyyn	$3\frac{1}{2}$	$2\frac{3}{8}$	$4\frac{1}{8}$	$3\frac{1}{2}$

Kadeesha would like to create a histogram of fabric amounts. Select all of the intervals for the bins that are reasonable.



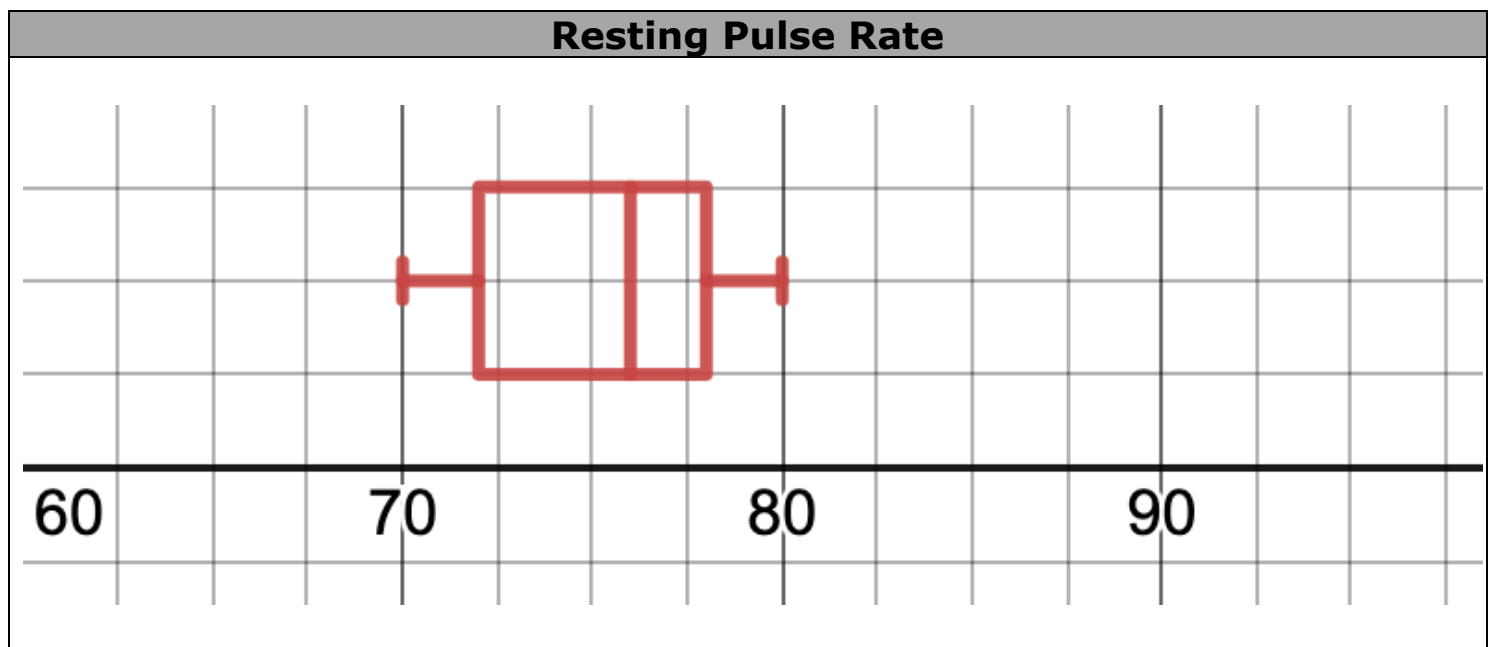
4. Terri recorded her resting pulse rate at different times of the day. Her data is shown and is labeled as data set A .

$$A = \{70, 70, 70, 70, 72, 72, 72, 73, 76, 76, 77, 77, 78, 78, 78, 80, 80, 80\}$$

- a. Determine each value. Show your work.

Statistical Measure	Work	Value
Median		76
Interquartile Range		6

- b. Create a box plot.



Terri added two measurements of her pulse rate during exercise to create the new data set B , as shown.

$$B = \{70, 70, 70, 70, 72, 72, 72, 73, 76, 76, 77, 77, 78, 78, 78, 80, 80, 80, 120, 142\}$$

The mean of data set A is $74.9\bar{4}$. The mean of data set B is 80.55.

c. Complete the statement.

The measure of center that should be used for data set B is the

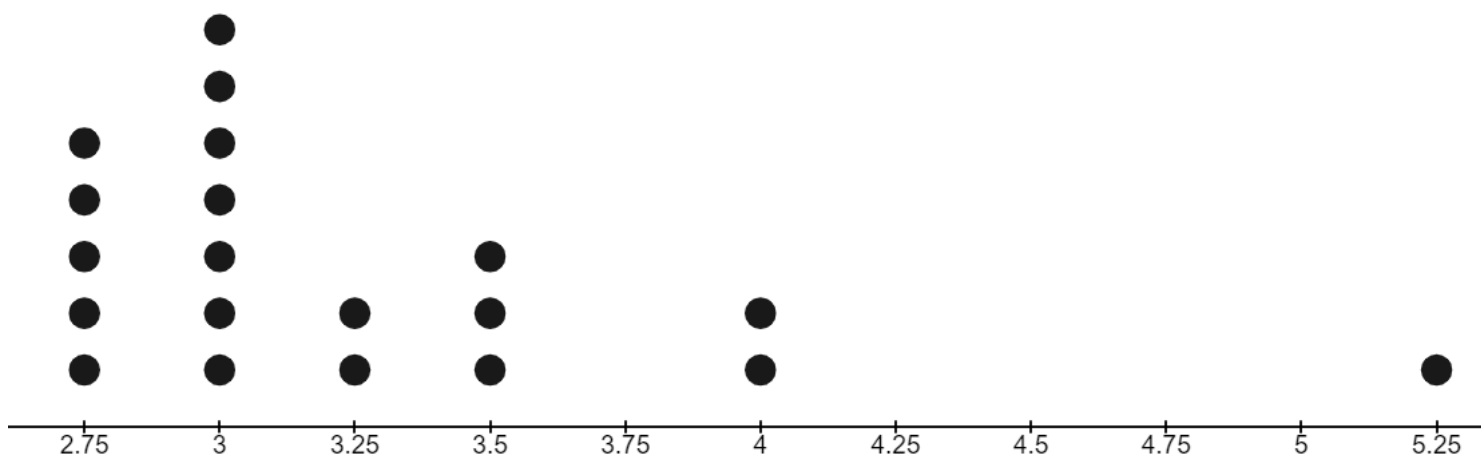
- ☐ mean
- ☒ median

because the two additional pulse rates that were added

to data set A to create data set B influenced the

- ☒ range.
- ☐ interquartile range.

5. The cost, in dollars, of a healthy meal in 20 different countries is shown in the dot plot (line plot).



- a. Which statistical question may result in the data shown in the dot plot (line plot)?

A. What is the cost, in dollars, of a healthy meal?

B. How many countries have a healthy meal that costs \$3.50?

C. Which countries have healthy meals that cost less than \$5.50?

D. How much, per person, does it cost to make a meal in the United States?

- b. Describe the shape of the dot plot (line plot), including any gaps, outliers, or clusters.

The shape of the graph is skewed. Data points are clustered around the values 2.75 to 3.5. There is a gap between 3.5 and 4 and another gap between 4 and 5.25. The data point 5.25 is an outlier.